

Mingqian Zhou

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EDUCATION

University of Cambridge

Cambridge, United Kingdom

Master of Philosophy in Advanced Computer Science (Expected: Distinction)

Sep. 2025 – Jun. 2026

- Coursework: Machine Visual Perception, Affective AI, Practical Research in Human-Centered AI, Computing for Collective Intelligence, Geometric Deep Learning
- Research Interests: Embodied AI, Humanoid Whole-Body Control, Sim-to-Real Transfer, Reinforcement Learning

Northwestern Polytechnical University (NWPU)

Xi'an, China

Bachelor of Engineering in Robotics

Sep. 2021 – Jun. 2025

- Average score: 91.36/100 (**Rank: 1/345**)
- **Scholarships: National Scholarship** (Top5/345, 2024), **Model Student Pacesetter** (Top 10 University-wide, Highest Honor for Undergraduates), Outstanding Graduate of NWPU (Top 1%, 2025) “Su Zhou Yu Cai” Special Scholarship (Highest Award in NWPU, 2023), First Class Scholarship of NWPU (4 Times)
- **Grade of Major courses:** Theoretical Mechanics: 97, Calculus(advanced): 100, Control Element and Robot System: 95, Robot motion and control technology: 99, Introduction to robotics: 93

Khalifa University (Exchange)

Abu Dhabi, UAE

Major in Electrical Engineering

Aug. 2023 – Dec. 2023

- **GPA:** 4.0/4.0 (QS183/1st in UAE) | Exchange student selection ratio: 8/50
- **Courses:** Feedback control system (4.0), Electronic Circuits & Devices (4.0)

PUBLICATIONS / PREPRINTS

GEAR: Gated Evidence-Aware Routing for Functional 3D Scene Graph Grounding

M. Zhou and P. Lió.

Manuscript in preparation (target: IEEE RA-L).

Selective graph-verification framework for functional 3D scene-graph grounding under same-label object ambiguity.

Beyond Retargeting: Robot-Centric Video Hallucination for Zero-Shot Humanoid Interaction (Dream2Act)

W. Xu, J. Li, Y. Gu, B. Yang, H. Chen, S. Lin, M. Zhou, J. Tan, Q. Wu, X. Jiang, T. Wang, J. Wen, Q. Liang, J. Zhang, C. Fang, and R. Xu.

Conference on Robot Learning (CoRL), 2026 – Under Review.

Zero-shot humanoid interaction via robot-centric video hallucination and real-world Unitree G1 deployment.

RESEARCH EXPERIENCE

Dream2Act: Robot-Centric Video Hallucination for Humanoid Interaction

Guangzhou, China

Research Assistant (Prof. Renjing Xu)

Jan.–Sep. 2025

- Led real-world deployment and validation for Dream2Act (CoRL 2026, under review) on a full-sized Unitree G1, executing zero-shot whole-body tasks from monocular input with 70% sitting and 60% kicking success.
- Owned the hardware experiment protocol, including scene setup, safety checks, reset procedures, trial logging, and repeatable success-rate measurement on physical runs.
- Contributed to sim-to-real evaluation by testing robot-centric hallucinated trajectories against human-centric retargeting baselines and documenting spatial-contact failures.

GEAR: Functional 3D Scene Graph Grounding

Cambridge, UK

Graduate Researcher (Supervisor: Prof. Pietro Lió)

Ongoing

- Independently designed GEAR's candidate-conditioned evidence packets, structured graph prior, and validation-frozen router for selective LLM verification.

- Built FunGraph-Bench over SceneFun3D and 3DSSG with 1,412 frozen test queries, adding target, anchor, supporting-edge, edge-family, and same-label-distractor annotations.
- Matched or exceeded graph-conditioned LLM cascades while invoking the verifier on only 58.1% of queries; validated the deployable router at 0.7075 overall R@1 with 14.7% API calls.

Animating Dynamic 3D Scenes from a Single Image

Cambridge, UK

Graduate Researcher (Supervisor: Prof. Cengiz Öztireli)

Oct. 2025 – Jan. 2026

- Explored single-image lifting into dynamic 4D scene representations, comparing implicit NeRF-style backbones with explicit 3D Gaussian Splatting for view synthesis quality.
- Integrated video diffusion priors to distill text-guided temporal dynamics into static 3D assets while preserving geometric consistency.
- Designed a CLIP-based evaluation and ablation pipeline for semantic alignment, motion realism, and texture fidelity under common generative artifacts.

Generalist Policies for Cooperative Multi-Agent RL

Cambridge, UK

Graduate Researcher (Supervisor: Prof. Amanda Prorok)

Ongoing

- Developing a foundation-model-inspired generalist policy that learns reusable cooperative primitives for zero-shot transfer across unseen multi-agent tasks.
- Designing graph-based attentional communication modules for scalable decentralized execution and training multi-task curricula in VMAS.
- Benchmarking transfer and robustness in BenchMARL against MARL baselines such as MAPPO.

Dynamic Reactive Motion Planning of Bipedal Robots

Xi'an, China

Research Assistant (Supervisor: Prof. Zhaohui Yuan)

Apr. 2023 – Jun. 2024

- Developed a hierarchical MPC/WBC framework for stable bipedal locomotion under external disturbances and unknown-environment constraints.
- Proposed a Linear Reaction Wheel Pendulum MPC for online gait generation and velocity tracking.
- Validated whole-body dynamic simulations and NAO hardware experiments using centroidal-Jacobian inverse kinematics and inverse dynamics.

Robust Perception and Navigation for Rescue Robot

Xi'an, China

Project Leader (Supervisor: Prof. Wentao Jiang)

Jun. 2022 – Dec. 2023

- Led an autonomous navigation system for GPS-denied rescue settings, integrating Cartographer SLAM, A*/DWA planning, and semantic perception.
- Improved localization accuracy by 30% through scan-matching parameter tuning and fused monocular semantic labels into LiDAR maps for obstacle awareness.

AWARDS & HONOURS

Champion , 2022 China Robot Competition (RoboCup China)	Mar. 2022
International First Prize , Asia-Pacific Mathematical Contest in Modeling (MCM/ICM)	Jul. 2022
Outstanding Winner , Elite league of the National 3D competition	Aug. 2023
National First Prize , 2023 China Robot Competition (RoboCup China)	Oct. 2023
International First Prize , Mathematical Contest in Modeling (Top 5% worldwide)	Sep. 2023
National First Prize , Chinese National Mathematical Modeling Competition	Nov. 2023
National-level Undergraduate Innovation and Entrepreneurship Project Funding	2022, 2023
Star of Erudition (Top 1/345)	2023
Merit Student of NWPU (Top 5%)	2021, 2022, 2023, 2024

EXTRACURRICULAR ACTIVITIES

President , Student Union, NWPU – led university-wide academic and cultural events for 30,000+ students	Sep. 2021 – Jun. 2025
Vice President , Cambridge Industrialization Society – connected scholars and practitioners through conferences, seminars, and industry events	Feb. 2026 – Now

SKILLS

Languages: English: Fluent (IELTS: 8.0), Mandarin: Native

Programming & Tools: Python (PyTorch), C++, MATLAB, L^AT_EX, Isaac Gym, ROS